

Identifying Determinants of Picture Naming Latencies in a Large-scale Picture-Word Interference Dataset

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Motivation: Semantic Similarity in PWI



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RQ1: Predictor Discovery

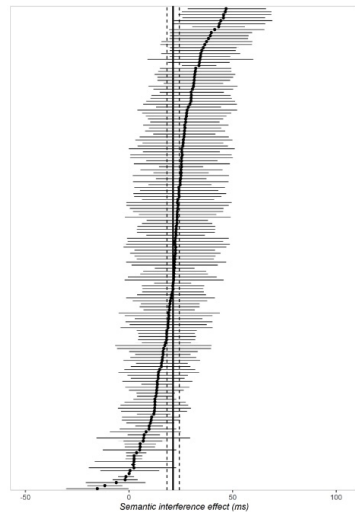
Which word properties and experimental design features are most important for predicting reaction times?

- **Goal:** Identify the most influential predictors among 39 word and design variables.
- Application: 80% training partition of **655,254** trials

Motivation: Semantic Similarity in PWI

PWI Literature Challenges

- **Binary / categorical operationalizations**
 - ▶ Cohyponymy: Standard operationalization in PWI (Bürki et al., 2020; Korko et al., 2024)
 - ▶ High variability in effect sizes and conflicting findings (Abdel Rahman & Melinger, 2019; Bürki et al., 2020)
- **Continuous semantic operationalizations**
 - ▶ Many semantic relations might be better captured by continuous measures (Günther et al., 2019)
 - ▶ Yet semantic similarity remains largely unexplored (Bürki et al., 2020; Korko et al., 2024)

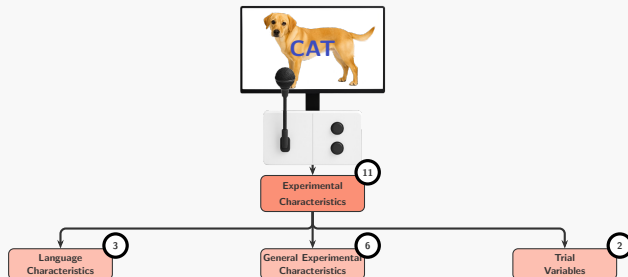


Interference effect of 21 ms (Bürki et al., 2020)

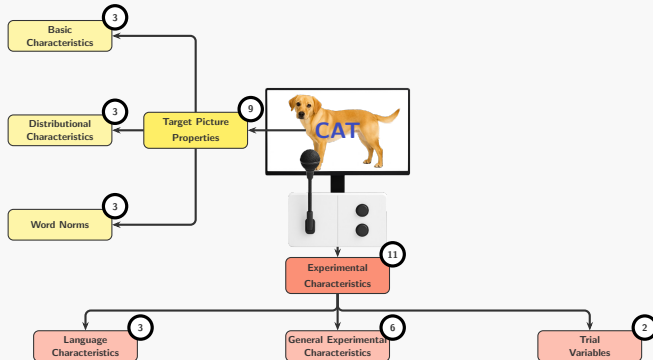
Overview of 39 Predictors



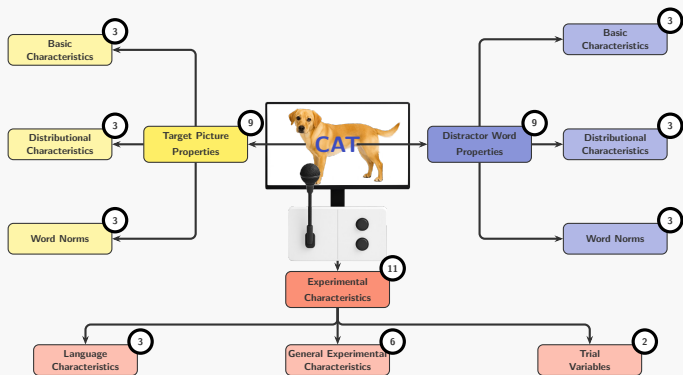
Overview of 39 Predictors



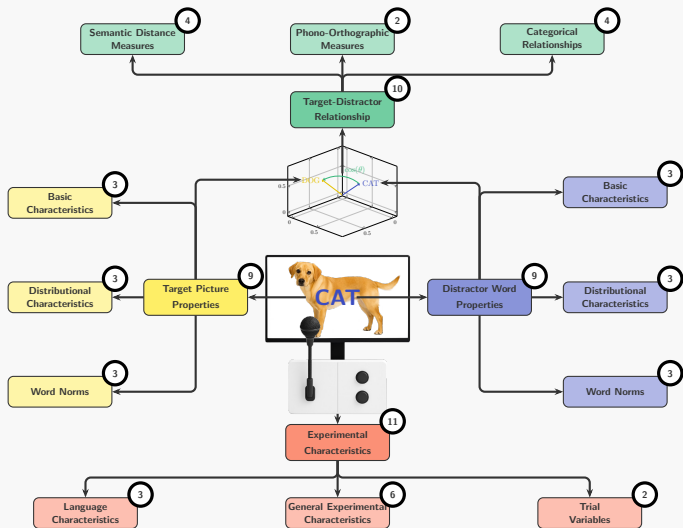
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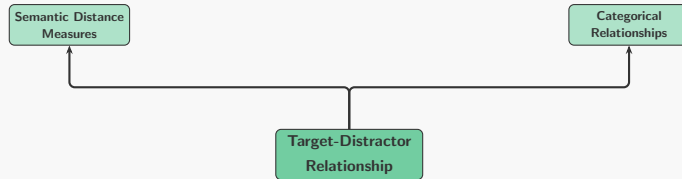
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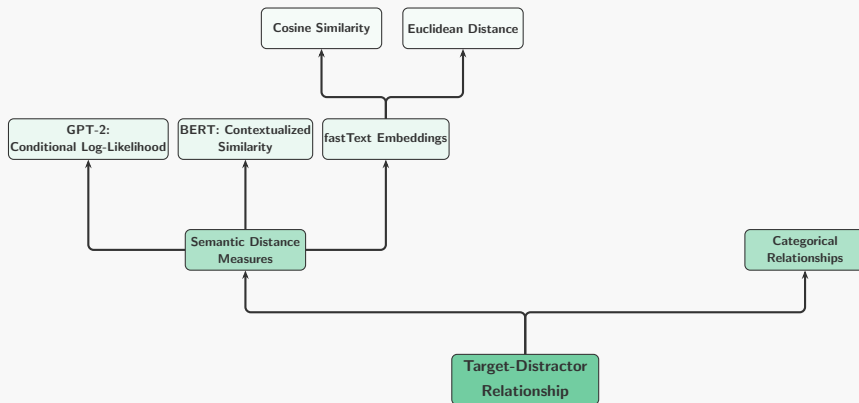
Overview of 39 Predictors



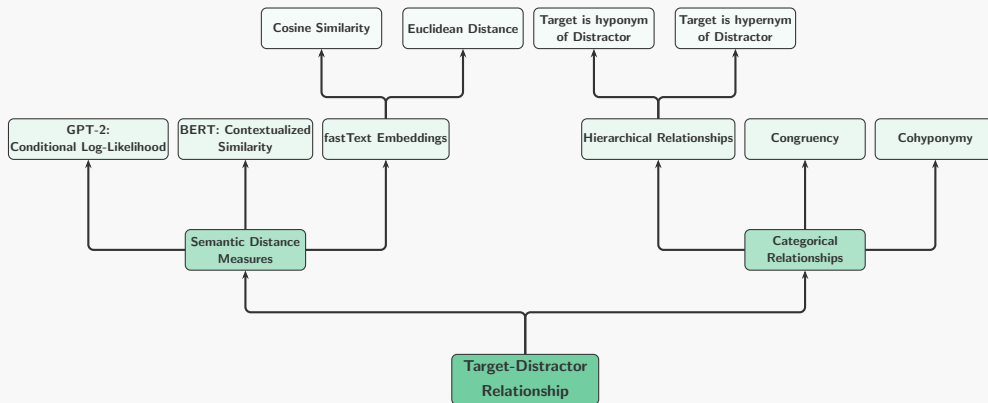
Semantic Variables



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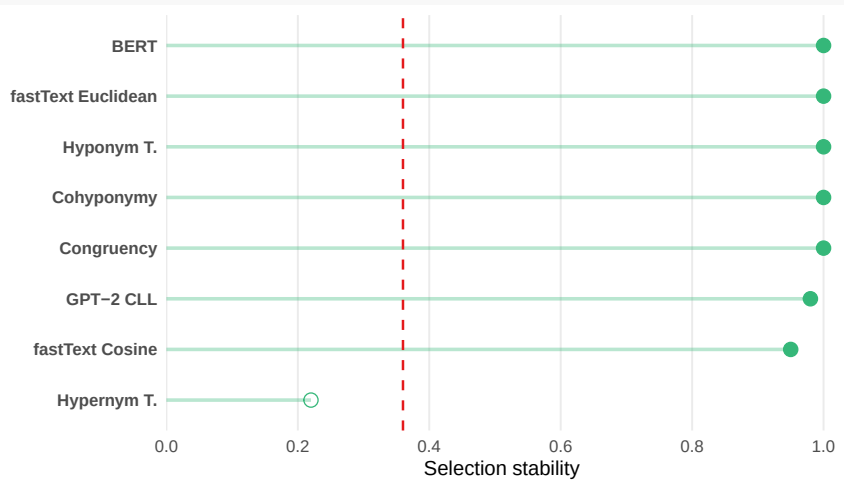


Model Specification

$$\log(RT_i) = \sum_{k=1}^p x_{ik}\beta_k + u_{j(i)}^{\text{exp}} + u_{k(i)}^{\text{part}} + u_{l(i)}^{\text{targ}} + u_{m(i)}^{\text{dist}} + \epsilon_i$$

- $\log(RT_i)$: Log-transformed reaction time for observation i
- $\mathbf{x}_i = (x_{i1}, \dots, x_{ip})$: Vector of $p = 39$ fixed-effect predictors for observation i
- $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)$: Vector of fixed-effect coefficients
- Random intercepts for observation i :
 - ▶ $u_{j(i)}^{\text{exp}}$: Experiment
 - ▶ $u_{k(i)}^{\text{part}}$: Participant
 - ▶ $u_{l(i)}^{\text{targ}}$: Target
 - ▶ $u_{m(i)}^{\text{dist}}$: Distractor
- ϵ_i : Residual error term for observation i

Variable Selection Results: Stepwise Backward Selection + Stability Selection



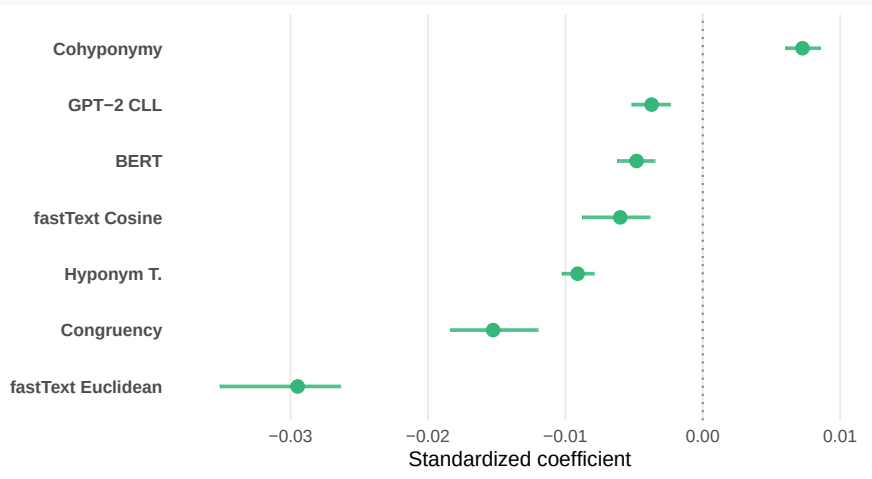
Stability Selection

- 100 iterations

Predictors Kept

- $39 \rightarrow 28$
- Data-driven threshold $t = 0.36$

Bootstrap Model Results



28-Variable-Model R^2

- $R^2_{\text{marginal}} = .227$
- $R^2_{\text{conditional}} = .636$

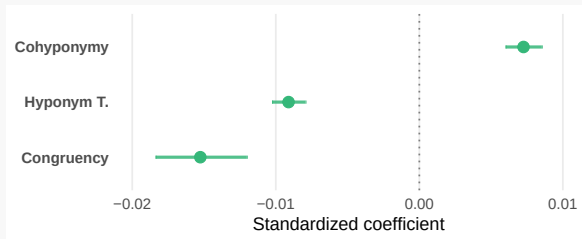
Interpretation

- All 28 retained predictors have a 95% CI excluding 0

Target–Distractor Relationships: Categorical & Hierarchical Relations

Categorical Relations

- **Cohyponymy**
 - ▶ **Interference**: as predicted (Bürki et al., 2020)
- **Congruency**
 - ▶ **Facilitation** as expected
- **Target is Hyponym of Distractor**
 - ▶ **Facilitation**: Other studies found no effect/facilitation Korko et al. (2024)



Target–Distractor Relationships: Similarity & Predictability

Contextual Predictability

- **GPT-2 Conditional Log-Likelihood**

- ▶ **Facilitation**: Distractors more likely given the target → shorter RTs

Semantic Similarity & Distance

- **BERT Contextualized Similarity**

- ▶ **Facilitation**: Semantically similar TD-pairs → shorter RTs

- **fastText Cosine Similarity**

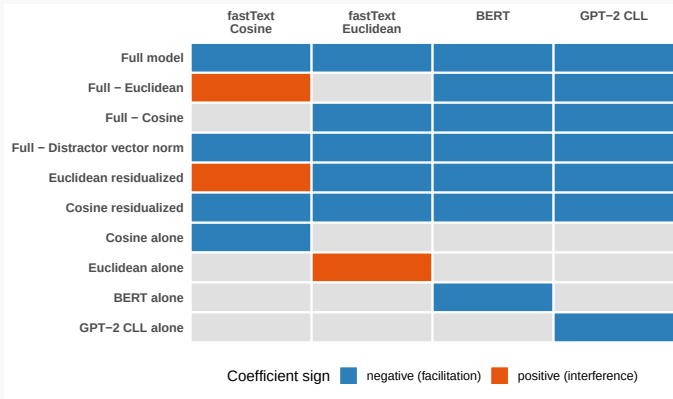
- ▶ **Facilitation**: Semantically similar TD-pairs → shorter RTs

- **fastText Euclidean Distance**

- ▶ **Interference**: Semantically distant TD-pairs → shorter RTs



Exploratory: Sign Stability Across Specifications



Sign (in)stability

- **BERT** and **GPT-2 CLL** stay negative (facilitative) across *all* specifications
- **fastText Cosine** and **Euclidean** flip sign when the partner measure or vector norms are removed/residualized
- $\Rightarrow$ the fastText measures must be read *jointly*, not individually

Discussion: Implications for Semantic Effects in PWI

From binary categories to continuous semantic similarity

- Continuous, model-derived semantic similarities capture target–distractor structure that 0/1 categorical/associative codes blur (e.g. *backpack–school*)
- **Categorical relations still inform** – congruency facilitates, cohyponymy interferes, target-hyponymy facilitates – but continuous measures add resolution

Which continuous measure can we trust?

- **fastText Cosine & Euclidean read jointly:** collinear and individually sign-unstable – together they index semantic similarity, but do not establish which operationalization carries the effect
- **BERT Contextualized Similarity** and **GPT-2 Conditional Log-Likelihood** carry a stable, facilitative sign across specifications
- *Implication:* do not neglect continuous, contextualized semantic similarities in PWI design and analysis!

References

-  Abdel Rahman, R., & Melinger, A. (2019). **Semantic processing during language production: An update of the swinging lexical network.** *Language, Cognition and Neuroscience*, 34(9), 1176–1192.
-  Bürki, A., Elbuy, S., Madec, S., & Vasishth, S. (2020). **What did we learn from forty years of research on semantic interference? A Bayesian meta-analysis.** *Journal of Memory and Language*, 114, 104125.
-  Günther, F., Rinaldi, L., & Marelli, M. (2019). **Vector-space models of semantic representation from a cognitive perspective: A discussion of common misconceptions.** *Perspectives on Psychological Science*, 14(6), 1006–1033.

-  Korko, M., Bose, A., Jones, A., Coulson, M., & de Mornay Davies, P. (2024). **Do words compete as we speak? A systematic review of picture-word interference (PWI) studies investigating the nature of lexical selection.** *Psychology of Language and Communication*, 261–322. <https://doi.org/10.58734/plc-2024-0011>
-  Schiekiera, L., Abdel Rahman, R., Gruber, V., Bürki, A., Lorenz, A., Stark, K., & Günther, F. (2026). **A harmonized trial-level dataset of picture-word interference.** https://osf.io/preprints/psyarxiv/xp69t_v1